

Calendar

Combinatorics

MATH 31003-01 - Fall 2026

I will put all deadlines on the [Blackboard Calendar](#). Please check it regularly.

0.1 Key university dates

For official University dates, see the [Fall 2026 academic calendar](#) and the [Registrar's final-exam schedule](#).

Date	Event
Monday, August 17	Classes begin
Monday, September 7	Labor Day holiday
Tuesday, November 24–Friday, November 27	Thanksgiving holiday
Wednesday, December 9	Final exam: 8:00-10:00 a.m., Wednesday, December 9, 2026

0.2 Exams

Date	Assessment
Friday, September 18	Midterm 1
Friday, October 16 TENTATIVE	Midterm 2
Friday, November 13 TENTATIVE	Midterm 3
8:00-10:00 a.m., Wednesday, December 9, 2026	Exam 4 and mastery reassessment

0.3 Daily lecture log

The public daily log below contains the lectures currently released. Later planning entries remain in the instructor's private lecture log and do not appear in this navigation.

Wk	Lec
	L1: Set notation and basic counting
2026-08-17	Sets, cardinality, Cartesian products, and the sum rule.
	L2: Product rule and strings
2026-08-19	Product rule; strings with and without repetition.
	L3: More counting techniques
2026-08-21	Subtraction, division, permutations, combinations, and counting models.
	L4: Applications of combinations and combinations with repetition
2026-08-22	Lattice paths. Repeated-letter anagrams and multinomial coefficients. Combinations with repetition; s
	L5: Review of choice models, increasing sequences, and basic binomial expansions
2026-08-26	Choosing the correct model: does order matter and are repetitions allowed. A second bijection for com
	L6: Polynomials, paths, Pascal recursion and restricted lattice paths, intro to inclusion–e
2026-08-28	Multinomial formula and polynomial expansions Coefficient extraction from powers of multivariate pol
	L7: Catalan paths and the inclusion–exclusion principle
2026-08-30	Reflection argument and Catalan numbers. Inclusion–exclusion for two, three, and finitely many sets.
	L8: Applications of inclusion–exclusion
2026-09-02	Characteristic functions. Conclusion of the proof of the inclusion–exclusion principle. Complement and
	L9: Applications of inclusion–exclusion
2026-09-04	Applications to bounded integer solutions and strings without repeated letters that avoid or contain sp
	L10: Applications of inclusion–exclusion
2026-09-09	Review of counting strings that avoid specified substrings. The inclusion–exclusion formula when an in
	L11: Applications of inclusion–exclusion
2026-09-11	Strings that use every symbol, surjective functions, and Stirling numbers. Graphs without isolated vert
TENTATIVE Lecture topics and assessment dates may change.	

0.4 Week and unit overview

Weeks	Unit	Topic
1–4	Unit I: Enumeration	Counting models, bijections, double counting, and incl
4–5	TENTATIVE Unit II: Generating-function foundations	Formal power series and ordinary generating-function
6–8	TENTATIVE Unit III: Recurrences	State models, linear recurrences, recursive evaluation,
9–16	TENTATIVE Further topics	To be announced